

AILT Study 5 (#128442)

Author(s)

Chiara Longoni (Boston University) - chiara.longoni2@unibocconi.it
Stephanie Tully (University of Southern California) - tullyste@marshall.usc.edu
Gil Appel (George Washington University) - gappel@gwu.edu

Pre-registered on: 2023/04/11 - 05:39 PM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

We are examining whether algorithmic knowledge impacts propensity to use AI tools to complete tasks, and whether any such relationship is explained by perceptions of AI as being more magical.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent measure will be preference to use AI (vs. a human) to complete 5 different tasks (all on 7-point scales; 1 = definitely prefer the human, 4 = indifferent, 7 = definitely prefer the AI algorithm).

The mediator capturing beliefs of AI as magical will be measured with participants' level of agreement with the following statements: (1) Artificial intelligence seems magical. (2) Artificial intelligence seems to conjure answers out of thin air. (3) Artificial intelligence seems like wizardry. (4) Artificial intelligence seems supernatural. (5) Artificial intelligence seems irrational. (6) Artificial intelligence has an air of mystique. (7) If I could see how artificial intelligence works, it would look like a magic show (all on 7-point scales; 1 = strongly disagree, 7 = strongly agree).

4) How many and which conditions will participants be assigned to?

There will be no conditions. Our independent variable will be participants' algorithmic literacy as measured using a 25-item multiple choice test. The sum of the correct responses will serve as their knowledge level.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will calculate participants' preference to use AI across the 5 different tasks. We will then regress this average AI preference on their AI/algorithmic literacy score.

We will calculate participants' perception of AI as magical by averaging across the 7 measures. We will regress these perceptions on their AI/algorithmic literacy score.

We will examine whether perceptions of AI as more magical mediate the relationship between knowledge and preference for AI using Hayes' PROCESS macro 4.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We include one instructional manipulation check (IMC; "Please indicate you are taking the time to read the instructions carefully by completing the following simple math question. If Jennie starts with a dozen eggs and uses four eggs to make an omelette, how many eggs does she have left? Please write a numeric answer only (no letters). Answering this question incorrectly will end the survey."). This will be asked at the beginning of the survey and participants who fail this question will not complete the survey.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will aim to recruit 1,000 participants.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will collect demographic information (gender, age, marital status, employment status and industry).